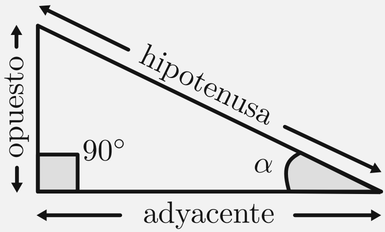
TRIGONOMETRÍA

— YisusReveck —



S	$\frac{o}{h}$	$\text{sen } \alpha = \dfrac{\text{opuesto}}{\text{hipotenusa}}$
C	$\frac{a}{h}$	$\cos \alpha = \dfrac{\text{adyacente}}{\text{hipotenusa}}$
T	$\frac{o}{a}$	$\tan \alpha = \dfrac{\text{opuesto}}{\text{adyacente}}$

$\measuredangle\alpha$ grados	π radianes	sen α	cos α	tan α
$0°$	0	0	1	0
$30°$	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$45°$	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$60°$	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90°$	$\frac{\pi}{2}$	1	0	—
$180°$	π	0	-1	0
$270°$	$\frac{3\pi}{2}$	-1	0	—

IDENTIDADES TRIGONOMÉTRICAS

— YisusReveck —

Recíprocas

$$\text{sen } \alpha = \frac{1}{\csc \alpha} \qquad \csc \alpha = \frac{1}{\text{sen } \alpha}$$

$$\cos \alpha = \frac{1}{\sec \alpha} \qquad \sec \alpha = \frac{1}{\cos \alpha}$$

$$\tan \alpha = \frac{1}{\cot \alpha} \qquad \cot \alpha = \frac{1}{\tan \alpha}$$

Pitagóricas

$$\text{sen}^2\alpha + \cos^2\alpha = 1$$

$$1 + \tan^2\alpha = \sec^2\alpha$$

$$1 + \cot^2\alpha = \csc^2\alpha$$

Cociente

$$\tan \alpha = \frac{\text{sen } \alpha}{\cos \alpha} \qquad \cot \alpha = \frac{\cos \alpha}{\text{sen } \alpha}$$

Ángulos opuestos

$$\text{sen}(-\alpha) = -\text{sen } \alpha$$

$$\cos(-\alpha) = \cos \alpha$$

$$\tan(-\alpha) = -\tan \alpha$$

Ángulos dobles

$$\text{sen}(2\alpha) = 2\,\text{sen } \alpha \cos \alpha$$

$$\cos(2\alpha) = \cos^2\alpha - \text{sen}^2\alpha$$

$$\tan(2\alpha) = \frac{2\tan \alpha}{1 - \tan^2\alpha}$$

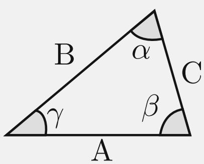
Suma y resta

$$\text{sen}(\alpha \pm \beta) = \text{sen } \alpha \cos \beta \pm \cos \alpha \text{ sen } \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \text{sen } \alpha \text{ sen } \beta$$

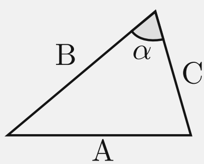
$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Ley de senos



$$\frac{A}{\text{sen } \alpha} = \frac{B}{\text{sen } \beta} = \frac{C}{\text{sen } \gamma}$$

Ley de cosenos



$$A^2 = B^2 + C^2 - 2\,BC \cos \alpha$$

INTEGRALES

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$$\int dx = x + C$$

$$\int k\,dx = kx + C$$

$$\int (u \pm v)\,dx = \int u\,dx \pm \int v\,dx$$

$$\int k\,f(x)\,dx = k\int f(x)\,dx$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C; \qquad n \neq -1$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\ln a} + C; \qquad a > 0,\ a \neq 1$$

$$\int \ln x\,dx = x\ln x - x + C$$

$$\int \operatorname{sen} x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \operatorname{sen} x + C$$

$$\int \tan x\,dx = -\ln|\cos x| + C$$

$$\int \cot x\,dx = \ln|\operatorname{sen} x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x\,dx = \ln|\csc x - \cot x| + C$$

$$\int \sec x\,\tan x\,dx = \sec x + C$$

$$\int \csc x\,\cot x\,dx = -\csc x + C$$

$$\int \operatorname{sen}^2 x\,dx = \frac{1}{2}x - \frac{1}{4}\operatorname{sen}(2x) + C$$

$$\int \cos^2 x\,dx = \frac{1}{2}x + \frac{1}{4}\operatorname{sen}(2x) + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \operatorname{arcsen}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln|x + \sqrt{x^2 - a^2}| + C$$

$$\int \frac{x}{a^2 + x^2}\,dx = \frac{1}{2}\ln\left(a^2 + x^2\right) + C$$

$$\int u\,dv = uv - \int v\,du$$

L	=	Logarítmicas
I	=	Inversas
A	=	Algebraicas
T	=	Trigonométricas
E	=	Exponenciales

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

DERIVADAS

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$$\frac{d}{dx}(k) = 0; \qquad k \text{ es constante}$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(k\,u) = k\,u'$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(u^n) = nu^{n-1}\cdot u'$$

$$\frac{d}{dx}(\sqrt{u}) = \frac{u'}{2\sqrt{u}}$$

$$\frac{d}{dx}(\sqrt[n]{u}) = \frac{u'}{n\sqrt[n]{u^{n-1}}}$$

$$\frac{d}{dx}\left(\frac{1}{u}\right) = -\frac{u'}{u^2}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$\frac{d}{dx}(u \pm v) = u' \pm v'$$

$$\frac{d}{dx}(u\,v) = u'v + uv'$$

$$\frac{d}{dx}(e^u) = e^u\cdot u'$$

$$\frac{d}{dx}(a^u) = a^u \ln a \cdot u'$$

$$\frac{d}{dx}(u^v) = u^v\left(v'\ln u + v\frac{u'}{u}\right)$$

$$\frac{d}{dx}(\ln u) = \frac{u'}{u}$$

$$\frac{d}{dx}(\log_a u) = \frac{u'}{u\ln a}$$

$$\frac{d}{dx}(\operatorname{sen} u) = \cos u \cdot u'$$

$$\frac{d}{dx}(\cos u) = -\operatorname{sen} u \cdot u'$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \cdot u'$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \cdot u'$$

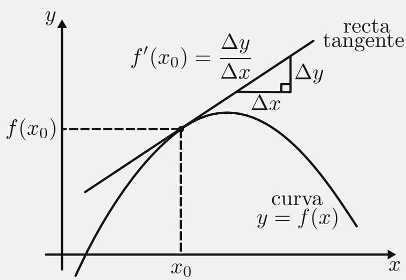
$$\frac{d}{dx}(\sec u) = \sec u \tan u \cdot u'$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \cdot u'$$

$$\frac{d}{dx}(\operatorname{arcsen} u) = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arccos u) = -\frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arctan u) = \frac{u'}{1+u^2}$$



$$\frac{d}{dx}f(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x))\cdot g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

GEOMETRÍA

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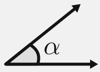
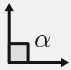
Agudo



$0 < \alpha < 90°$

Recto

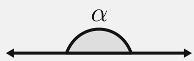


$\alpha = 90°$

Obtuso

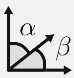
$90 < \alpha < 180°$

Llano



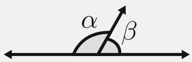
$\alpha = 180°$

Complementarios



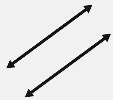
$\alpha + \beta = 90°$

Suplementarios

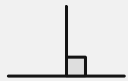


$\alpha + \beta = 180°$

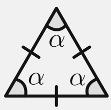
Paralelas



Perpendiculares

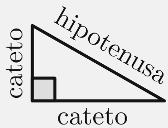


△ Equilátero

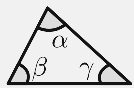


△ Isósceles

△ Rectángulo



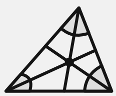
△ Escaleno



$\alpha \neq \beta \neq \gamma$

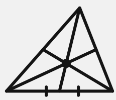
Líneas notables del triángulo

Bisectriz



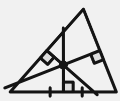
Incentro

Mediana



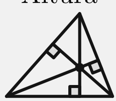
Baricentro

Mediatriz



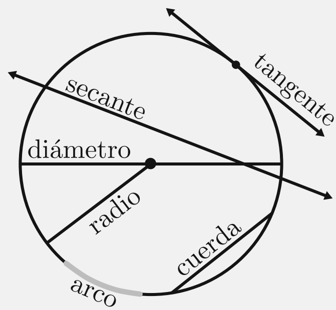
Circuncentro

Altura

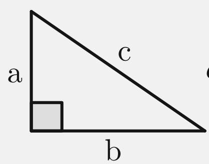


Ortocentro

Partes del círculo

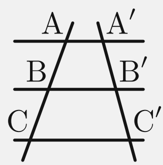


Teorema de Pitágoras



$$a^2 + b^2 = c^2$$

Teorema de Tales



$$\frac{AB}{A'B'} = \frac{BC}{B'C'}$$

ÁLGEBRA

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$$a^0 = 1 \qquad a^{-n} = \frac{1}{a^n}$$

$$\frac{a^m}{a^n} = a^{m-n} \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$a^m \cdot a^n = a^{m+n} \qquad \sqrt[m]{a}\,\sqrt[n]{a} = a^{\frac{1}{m}+\frac{1}{n}}$$

$$(ab)^n = a^n b^n \qquad \sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \qquad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(a^m)^n = a^{mn} \qquad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

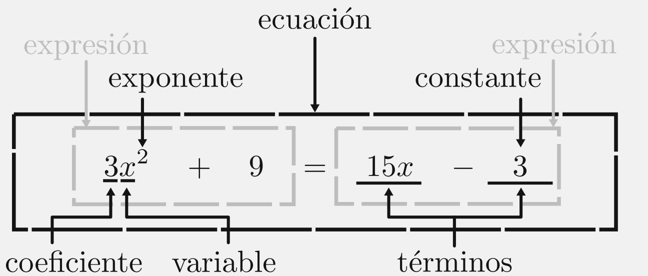
$$\log_b(x^k) = k\log_b x$$

$$\log_b x = \frac{\log_k x}{\log_k b}$$

$$\ln x = \log_e x$$

$$\ln(e^x) = x \qquad e^{\ln x} = x$$

$$\log_b 1 = 0 \qquad \log_b b = 1$$



π	$\approx$	3.14159
e	$\approx$	2.71828
$\sqrt{2}$	$\approx$	1.41421
φ	$\approx$	1.61803
τ	$\approx$	6.28318

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

$\Delta > 0$	2 soluciones reales
$\Delta = 0$	1 solución real
$\Delta < 0$	0 soluciones reales

$$(a+b)(a-b) = a^2 - b^2 \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (x+a)(x+b) = x^2 + (a+b)x + ab$$

FIGURAS

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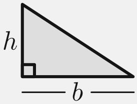
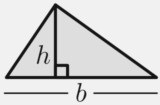
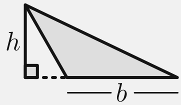
Triángulo

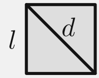






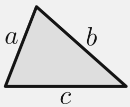
$$A = \frac{b \cdot h}{2}$$

Cuadrado



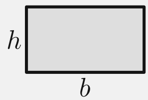
$$A = l^2$$
$$d = l \cdot \sqrt{2}$$

Fórmula de Herón



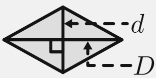
$$A = \sqrt{s(s-a)(s-b)(s-c)}$$
$$s = \frac{a+b+c}{2}$$

Rectángulo



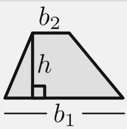
$$A = b \cdot h$$

Rombo



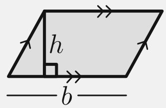
$$A = \frac{D \cdot d}{2}$$

Trapecio



$$A = (b_1 + b_2) \cdot \frac{h}{2}$$

Paralelogramo



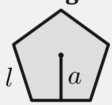
$$A = b \cdot h$$

Círculo

$$A = \pi r^2$$
$$P = 2\pi r$$

Polígonos regulares

Pentágono

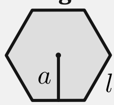


$$P = 5l$$

$$A = \frac{P \cdot a}{2}$$

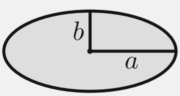
— a: apotema —

Hexágono



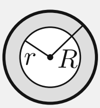
$$P = 6l$$

Elipse



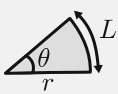
$$A = \pi a b$$

Corona circular



$$A = \pi(R^2 - r^2)$$
$$P = 2\pi(R + r)$$

Sector circular



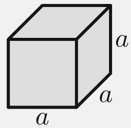
$$A = \frac{\theta r^2}{2}$$
$$L = r\theta$$

— θ en radianes —

CUERPOS

— YisusReveck —

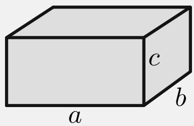
Cubo



$$A = 6a^2$$

$$V = a^3$$

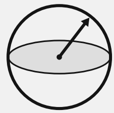
Ortoedro



$$A = 2(ab + ac + bc)$$

$$V = abc$$

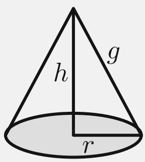
Esfera



$$A = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$

Cono

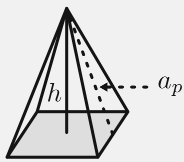


$$A_L = \pi r g$$

$$A_T = \pi r g + \pi r^2$$

$$V = \frac{1}{3}\pi r^2 h$$

Pirámide

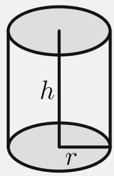


$$A_L = \frac{P_B \cdot a_p}{2}$$

$$A_T = A_L + A_B$$

$$V = \frac{1}{3}A_B \cdot h$$

Cilindro

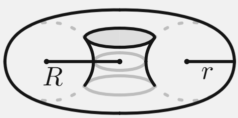


$$A_L = 2\pi r h$$

$$A_T = 2\pi r h + 2\pi r^2$$

$$V = \pi r^2 h$$

Toro (Dona)

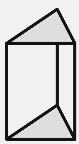


$$A = 4\pi^2 R r$$

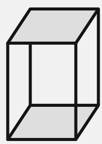
$$V = 2\pi^2 R r^2$$

Prismas rectos

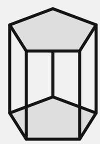
Triangular



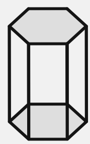
Cuadrangular



Pentagonal



Hexagonal



$A_L = P_B \cdot h$ $\quad A_T = A_L + 2A_B$ $\quad V = A_B \cdot h$

A_L = Área Lateral	A_T = Área Total	P_B = Perímetro de la Base
A_B = Área de la Base	g = Generatriz	a_p = Apotema de la pirámide